

Randall Preston

Alton, NH | randallpreston0123@gmail.com | (603) 507-2539 | www.linkedin.com/in/randall-preston

EDUCATION

Villanova University | College of Engineering | Villanova, PA **May 2029**
Bachelor of Science, Mechanical Engineering | Business Minor
Relevant Coursework: Engineering CAD/Programming, Statics, Circuits, Differential Equations, Calculus
GPA: 3.88/4.0 | Dean's List (Fall 2025-Present)

Summer Business Institute | Villanova University – Business Minor **Summer 2026**
16-Credit, 10-week intensive program focused on key concepts in accounting, business law, finance, economics, analytics, information technology, management, marketing, and professional development

ENGINEERING EXPERIENCE

NovaRacing FSAE | *Drivetrain System Design Engineer* | Villanova University **October 2025-Present**

VU18 Design Process **Spring 2026-Present**

- Individually overseeing iterative design process for drivetrain system
- Planning upgrade of half-shafts to composite design from steel resulting in 58% mass reduction
- Organizing additive manufacturing process for production of differential-engine mounts, allowing for increased mass/cost reduction and topological optimization of design

VU17 Design Process **Winter-Spring 2026**

- Participated in goal-oriented design, manufacturing, and assembly of drivetrain components
- Collaborated on inter-system work manufacturing 30+ carbon-fiber aerodynamics components for VU17 car and Senior Capstone Project

Undergraduate Research Scholar Program | Villanova University **March 2026-Present**

Hydrogen Permeation Testing in Structural Metals

- Conducts research with faculty advisor and team of six undergraduate/doctoral researchers on hydrogen permeation of structural metals for application in energy/automotive industries
- Designs, builds, and validates an electrochemical permeation testing setup based on the Devanathan-Stachurski dual-cell utilizing
- Meets weekly with team and advisor to maintain communication between different task forces and use research results to direct other experimentation parameters for the laboratory

LEADERSHIP & ACTIVITIES

Pi Kappa Alpha Fraternity | *Historian & Real Estate Chair* | Villanova University **April 2026-Present**

- Oversees meeting/event documentation, year-end report, and Petition to Install
- Organizes biweekly meetings with six-member real estate committee and alumni advisors

Villanova Band | *Trumpet Player* | Villanova University **September 2025-Present**

- Attends weekly rehearsals, plays at athletic/admission events

Scouting BSA | *Eagle Scout* | Troop 53, Alton, NH **2019-2026**

SKILLS

Design & Analysis | CAD (SolidWorks); MATLAB; ANSYS (Structural, CFD - Novice); Excel
Manufacturing | Carbon Fiber Composites (Wet Layup, Resin Infusion, Post-Processing, Assembly); GD&T; Machining (Manual Mill, Manual Lathe); Precision Measurement; Soldering; Welding - Novice